

Day 33 (Part 3): IPv6

CORRECTION TO PRIOR LECTURES:

RFC Requirements for IPv6 Address Representation

- Leading 0s MUST be removed
 - This - 2001 : 0db8 : 0000 : 0001 : 0f2a : 4fff : fea3 : 00b1
 - Becomes - 2001 : db8 : 0 : 1 : f2a : 4fff : fea3 : b1
- :: MUST be used to shorten the longest string of all-0 quartets
 - If there is only ONE all-0 quartet, don't use '::'
 - This - 2001 : 0000 : 0000 : 0000 : 0f2a : 0000 : 0000 : 00b1
 - Becomes - 2001 :: f2a : 0 : 0 : b1
- If there are two equal-length choices for the :: , use :: to the shorten the one on the LEFT
 - This - 2001 : 0db8 : 0000 : 0000 : 0f2a : 0000 : 0000 : 00b1
 - Becomes - 2001 : db8 :: f2a : 0 : 0 : b1
- Hexadecimal characters 'a', 'b', 'c', 'd', 'e', and 'f' MUST be written using lower-case, NOT upper case A B C D E F

IPv6 HEADER

Fixed header format																																	
Offsets	Octet	0								1								2								3							
Octet	Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	Version				Traffic Class								Flow Label																			
4	32	Payload Length																Next Header								Hop Limit							
8	64	Source Address																															
12	96																																
16	128																																
20	160																																
24	192	Destination Address																															
28	224																																
32	256																																
36	288																																

Length is ALWAYS 40 bytes (Fixed Header)

Version (4 bits)

- Indicates version of IP used
- Fixed value of '6' (0b0110) to indicate IPv6

Traffic Class (8 bits)

- Used for QoS (Quality of Service) to indicate high-priority traffic
- Example: IP phone traffic, live video calls, etc.

Flow Label (20 bits)

- Identifies specific traffic "flows" (communication between Source and Destination)

Payload Length (16 bits)

- Indicates the LENGTH of the PAYLOAD (the encapsulation LAYER 4 SEGMENT) **in bytes**
- The length of the IPv6 header, itself, isn't included, because it's ALWAYS 40 bytes

Next Header (8 bits)

- Indicates the TYPE of the 'next header' (header of the encapsulated SEGMENT)

- Example: TCP or UDP
- Same function as the IPv4 header's 'Protocol' field

Hop Limit (8 bits)

- Value in this field decrements by 1 every time a ROUTER forwards it. If it reaches '0', the PACKET is discarded (similar to IPv4 TTL field)

Source Address (128 bits)

- Packet's SOURCE address

Destination Address (128 bits)

- Packet's DESTINATION address

SOLICITED-NODE MULTICAST ADDRESS

- An IPv6 SOLICITED-NODE Multicast Address is calculated from a UNICAST ADDRESS

How to generate a SOLICITED-NODE Multicast Address

`ff02:0000:0000:0000:0000:0001:ff` + Last 6 hex digits of unicast address

`2001:0db8:0000:0001:0f2a:4fff:fea3:00b1`



`ff02::1:ffa3:b1`

Note the automatically joined group addresses for this IPv6 Interface

```

R1#sh ipv6 int g0/0
GigabitEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::EF8:22FF:FE36:8500
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8::EF8:22FF:FE36:8500, subnet is 2001:DB8::/64 [EUI]
Joined group address(es):
  FF02::1
  FF02::2
  FF02::1:FF36:8500
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 30000)
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.

```

NEIGHBOR DISCOVERY PROTOCOL (NDP)

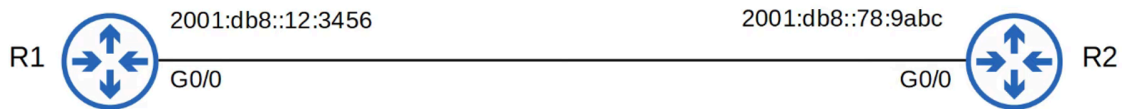
- NEIGHBOR DISCOVERY PROTOCOL (NDP) is a PROTOCOL used with IPv6
- It has various functions and one of those functions is to replace ARP, which is no longer used in IPv6
- The ARP-like function of NDP uses ICMPv6 and SOLICITED-MODE Multicast Addresses to learn the MAC ADDRESS of other HOSTS (ARP in IPv4 uses Broadcast Messages)
- TWO MESSAGES types are used:
 -
 - 1. NEIGHBOR SOLICITATION (NS)
 - ICMPv6 Type 135
 -

1. NEIGHBOR ADVERTISEMENT (NA)

- ICMPv6 Type 136



Neighbor Solicitation (NS)



Hi, what's your
MAC address?

- **Source IP:** R1 G0/0 IP
- **Destination IP:** R2 solicited-node multicast address
- **Source MAC:** R1 G0/0 MAC
- **Destination MAC:** Multicast MAC based on R2's solicited-node address

```
> Frame 6: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface -, id 0
> Ethernet II, Src: ca:01:09:6d:00:08 (ca:01:09:6d:00:08), Dst: IPv6mcast_ff:78:9a:bc (33:33:ff:78:9a:bc)
> Internet Protocol Version 6, Src: 2001:db8::12:3456, Dst: ff02::1:ff78:9abc
> Internet Control Message Protocol v6
```



Neighbor Advertisement (NA)



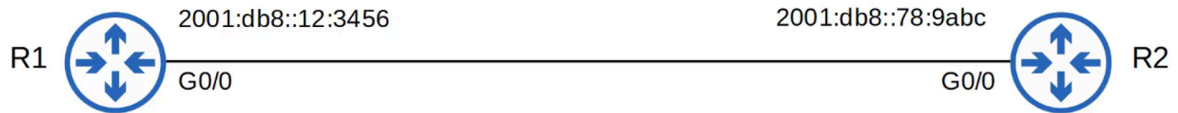
Hi, my MAC address is
ca02.097c.0008.

- **Source IP:** R2 G0/0 IP
- **Destination IP:** R1 G0/0 IP
- **Source MAC:** R2 G0/0 MAC
- **Destination MAC:** R1 G0/0 MAC

```
> Frame 7: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface -, id 0
> Ethernet II, Src: ca:02:09:7c:00:08 (ca:02:09:7c:00:08), Dst: ca:01:09:6d:00:08 (ca:01:09:6d:00:08)
> Internet Protocol Version 6, Src: 2001:db8::78:9abc, Dst: 2001:db8::12:3456
> Internet Control Message Protocol v6
```

IPv6 NEIGHBOR TABLE

IPv6 Neighbor Table



```
R1#show ipv6 neighbor
IPv6 Address                               Age Link-layer Addr State Interface
FE80::C802:9FF:FE7C:8                     0 ca02.097c.0008 REACH Gi0/0
2001:DB8::78:9ABC                         0 ca02.097c.0008 REACH Gi0/0
```

```
R2#show ipv6 neighbor
IPv6 Address                               Age Link-layer Addr State Interface
FE80::C801:9FF:FE6D:8                     0 ca01.096d.0008 REACH Gi0/0
2001:DB8::12:3456                         0 ca01.096d.0008 REACH Gi0/0
```

- Another function of NDP allows HOSTS to automatically discover ROUTERS on the LOCAL NETWORK
- TWO MESSAGES are used for this process:
 - ROUTER SOLICITATION (RS)
 - ICMPv6 Type 133
 - Sent to Multicast Address **FF02::2** (All Routers)
 - Asks ALL ROUTERS on the Local Link to identify themselves
 - Sent when an INTERFACE is enabled / HOST is connected to the NETWORK
 - ROUTER ADVERTISEMENT (RA)
 - ICMPv6 Type 134
 - Sent to Multicast Address **FF02::1** (All Nodes)
 - The ROUTER announces its presence, as well as other information about the link
 - These messages are sent in response to RS messages
 - They are also sent periodically, even if the ROUTER hasn't received an RS



SLAAC

- Stands for **STATELESS ADDRESS AUTO-CONFIGURATION**
- HOSTS use the RS / RA messages to learn the IPv6 Prefix of the LOCAL LINK (ie: 2000:db8:: /64) and then automatically generate an IPv6 Address
- Using the `ipv6 address prefix / prefix-length eui-64` command, you need to manually enter the prefix
- Using the `ipv6 address autoconfig` command, you DON'T need to enter the prefix. The device uses NDP to learn the prefix used on the local link
- The device will use EUI-64 to generate the INTERFACE ID or it will be randomly generated (depending on the device / maker)

```
R2(config)#int g0/0
R2(config-if)#ipv6 address autoconfig
R2(config-if)#do show ipv6 interface brief
GigabitEthernet0/0      [up/up]
    FE80::EF8:22FF:FE56:A600
    2001:DB8::EF8:22FF:FE56:A600
GigabitEthernet0/1      [administratively down/down]
    unassigned
GigabitEthernet0/2      [administratively down/down]
    unassigned
GigabitEthernet0/3      [administratively down/down]
    unassigned
```

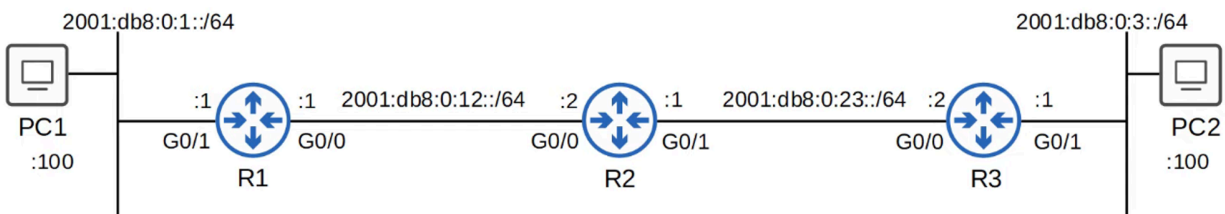
DUPLICATE ADDRESS DETECTION (DAD)

- One final point about NDP!

- Duplicate Address Detection (DAD) allows HOSTS to check if other devices on the Local Link are using the same IPv6 Address
- Any time an IPv6-enabled interface initializes (`no shutdown` command) or an IPv6 ADDRESS is configured on an INTERFACE (by any method: manual, SLAAC, etc.) it performs DAD
- DAD uses TWO MESSAGES you learned earlier : NS and NA
- The HOST will send an NS to its own IPv6 ADDRESS.
 - If it doesn't get a reply, it KNOWS the ADDRESS is unique
 - If it DOES get a reply, it means ANOTHER HOST on the NETWORK is already using that ADDRESS

IPv6 STATIC ROUTING

- IPv6 ROUTING works the same as IPv4 ROUTING
- However, the TWO processes are separate on the ROUTER, and the TWO routing tables are separate, as well.
- IPv4 ROUTING is enabled BY DEFAULT
- IPv6 ROUTING is disabled BY DEFAULT
 - MUST BE ENABLED with the `ipv6 unicast-routing` command
- If IPv6 ROUTING is disabled, the ROUTER will be able to SEND and RECEIVE IPv6 traffic, but will not *route* IPv6 traffic (ie: will NOT FORWARD it between NETWORKS)



```

R1#show ipv6 route
IPv6 Routing Table - default - 5 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       lA - LISP away, a - Application
C    2001:DB8:0:1::/64 [0/0]
    via GigabitEthernet0/1, directly connected
L    2001:DB8:0:1::1/128 [0/0]
    via GigabitEthernet0/1, receive
C    2001:DB8:0:12::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L    2001:DB8:0:12::1/128 [0/0]
    via GigabitEthernet0/0, receive
L    FF00::/8 [0/0]
    via Null0, receive

```

- A CONNECTED NETWORK ROUTE is automatically added for EACH CONNECTED NETWORK
- A LOCAL HOST ROUTE is automatically added for each ADDRESS configured on the ROUTER
- Routes for Link-Local ADDRESSES are not added to the ROUTING TABLE

```

ipv6 route destination/prefix-length {next-hop | exit-interface [next-hop]} [ad]

```

Everything is configured similar to normal static routes in IPv4

[AD] = Administrative Distance. You NEED this value in order to configure a STATIC ROUTE

DIRECTLY ATTACHED Static Route:

- Only the EXIT INTERFACE is specified
- `ipv6 route destination / prefix-length exit-interface`
- Example : `~~R1(config)# ipv6 route 2001:db8:0:3:: /64 g0/0~~`

💡 In IPv6, you CANNOT use DIRECTLY ATTACHED Static Routes if the INTERFACE is an ETHERNET INTERFACE

RECURSIVE Static Route:

- Only the Next-Hop is specified
- `ipv6 route destination / prefix-length next-hop`
- Example: `R1(config)# ipv6 route 2001:db8:0:3::/64 2001:db8:0:12::2`

FULLY SPECIFIED Static Route:

- Both the Exit Interface and Next Hop are specified
- `ipv6 route destination / prefix-length exit-interface next-hop`
- Example: `R1(config)# ipv6 route 2001:db8:0:3::/64 g0/0 2001:db8:0:12::2`

(NOTE THAT THESE ROUTES ARE ALL RECURSIVE : They specify the Next-Hop)

NETWORK ROUTE:

```
R1(config)# ipv6 route 2001:db8:0:3::/64 2001:db8:0:12::2
```

This is a route to R3/PC2 NETWORK via R2's G0/0 INTERFACE

(We did this in Day 32's Lab)

HOST ROUTE:

```
R2(config)# ipv6 route 2001:db8:0:1::100/128 2001:db8:0:12::1
```

```
R2(config)# ipv6 route 2001:db8:0:3::100/128 2001:db8:0:23::2
```

This is a route from R2 to PC1 and PC2 using the "next hop" ADDRESSES of R1 and R3 G0/0 INTERFACES

Note the /128 prefix. This is how SPECIFIC IPv6 ADDRESSES are written

DEFAULT ROUTE:

```
R3(config)# ipv6 route ::/0 2001:db8:0:23::1
```

::/0 is the IPv6 equivalent of 0.0.0.0/0 in IPv4

FLOATING STATIC ROUTES:

- Require you to increase the [AD] number HIGHER than the currently used NETWORK IGP AD value

LINK-LOCAL NEXT HOPS:

```

R1(config)#ipv6 route 2001:db8:0:3::/64 FE80::EF8:22FF:FEE6:D300
% Interface has to be specified for a link-local nexthop
R1(config)#
R1(config)#ipv6 route 2001:db8:0:3::/64 g0/0 FE80::EF8:22FF:FEE6:D300
R1(config)#do show ipv6 route
IPv6 Routing Table - default - 6 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
        B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
        H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
        IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
        ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
        RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
        OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
        LA - LISP away, a - Application
C    2001:DB8:0:1::/64 [0/0]
    via GigabitEthernet0/1, directly connected
L    2001:DB8:0:1::1/128 [0/0]
    via GigabitEthernet0/1, receive
S    2001:DB8:0:3::/64 [1/0]
    via FE80::EF8:22FF:FEE6:D300, GigabitEthernet0/0
C    2001:DB8:0:12::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L    2001:DB8:0:12::1/128 [0/0]
    via GigabitEthernet0/0, receive
L    FF00::/8 [0/0]
    via Null0, receive

```

You HAVE to specify the INTERFACE name when using Link-Local Next-Hops

This is EXACTLY like a FULLY-SPECIFIED STATIC ROUTE